HMI DESIGNING

KURVATUR FOLDER

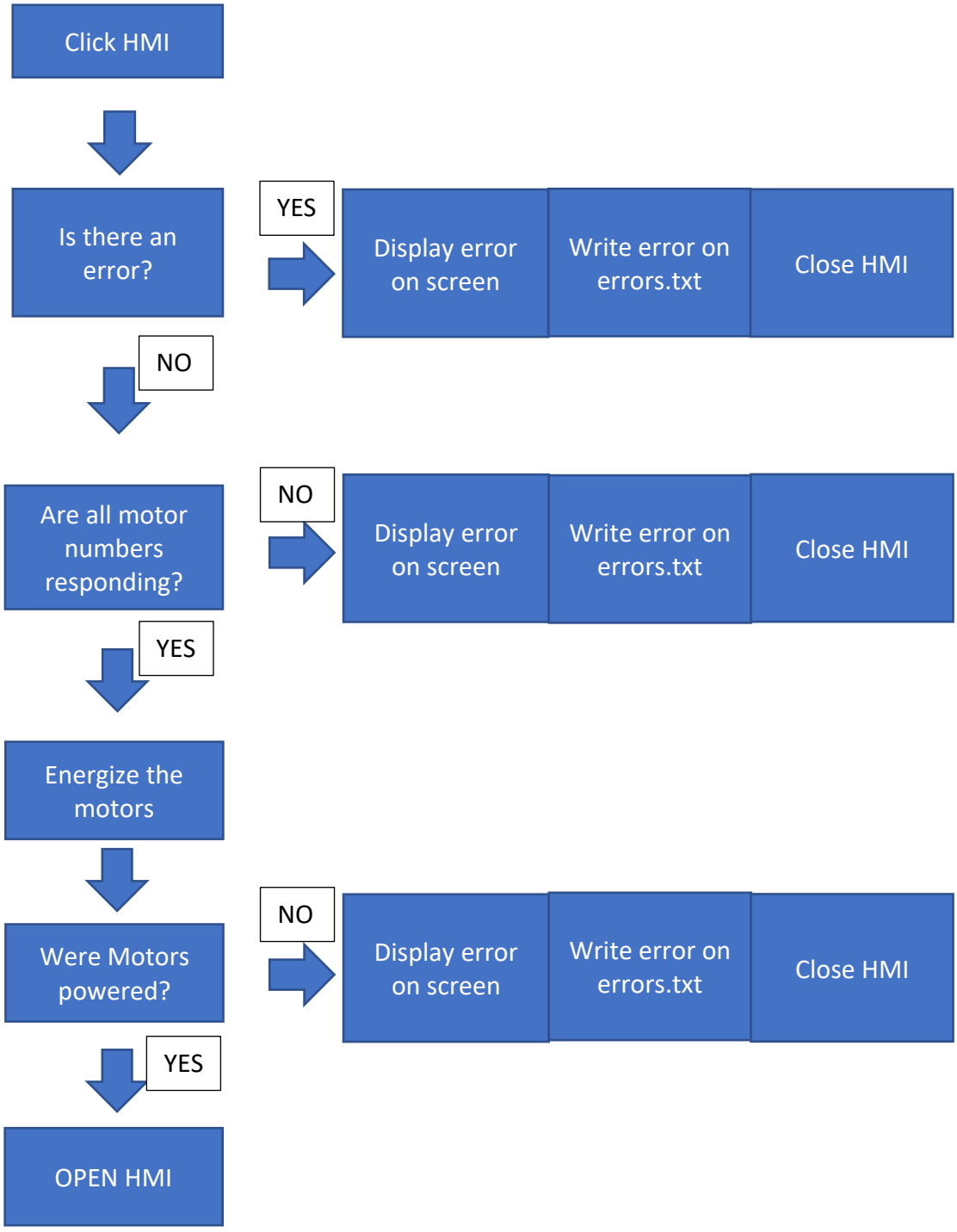
It is the folder named as “Kurvatur” on the Desktop. The folder content is shown below.

- **preferences (folder)**
  - **Calibration (folder)**
    - homePosition.txt
    - laserPosition.txt
  - **Status (folder)**
    - Date-1.txt
    - Date-2.txt
  - Motor.txt
  - MainBoard.txt
  - Errors.txt
  - Info.txt
- **Projects (folder)**
  - **Project A (folder)**
    - **Panel A (folder)**
      - Panel A.txt
      - **laserProjection**
        - Panel A.txt
        - Panel B.txt
        - Panel C.txt
    - **Panel B (folder)**
      - Panel B.txt
      - **laserProjection**
        - Laser 1.txt
        - Laser 2.txt
        - Laser 3.txt
    - **...**
  - **Project B (folder)**
    - **Part 1 (folder)**
      - Part 1.txt
      - **laserProjection**
        - Laser 1.txt
        - Laser 2.txt
        - Laser 3.txt
    - **Part 2 (folder)**
      - Part 2.txt
      - **laserProjection**
        - Laser 1.txt

OPENING HMI

Use: settings folder

- Check all the errors of motors and main board.
  If there is no error, going on opening HMI.
  If there is an error, display it on screen, write it preferences/errors.txt, and ask to close HMI
- Check all of motor numbers are responding
- Powering the motors. Check motors are powered.
- Open HMI

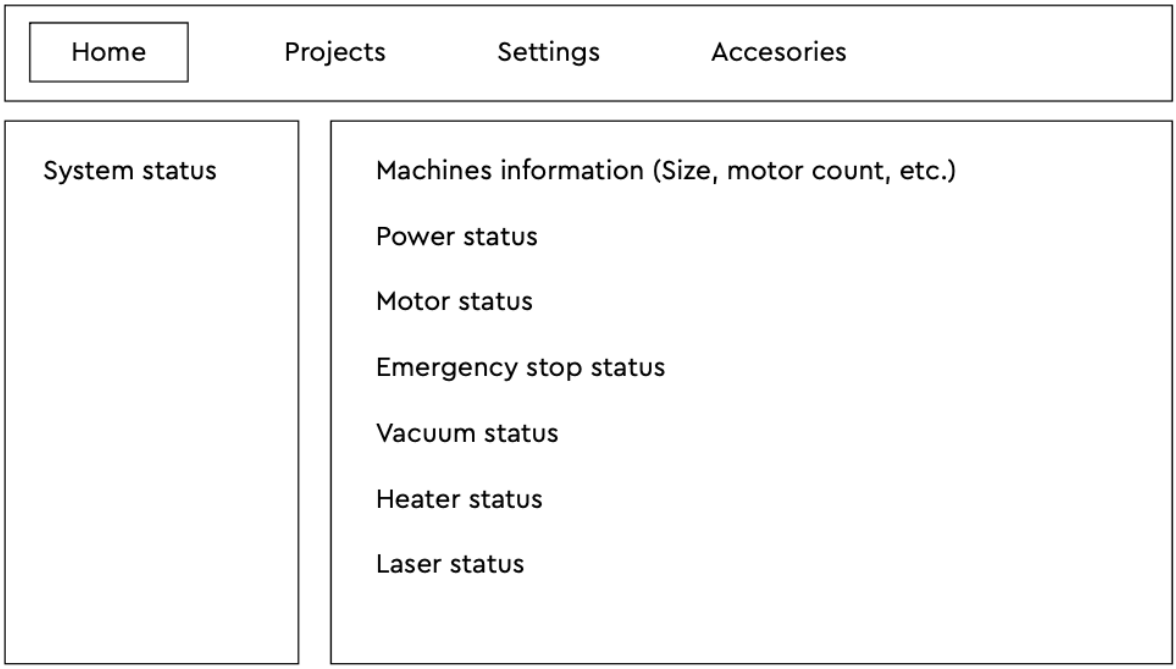


HMI CONTENT

1- HOME
2- PROJECTS
3- SETTINGS
4- ACCESSORIES

1- HOME

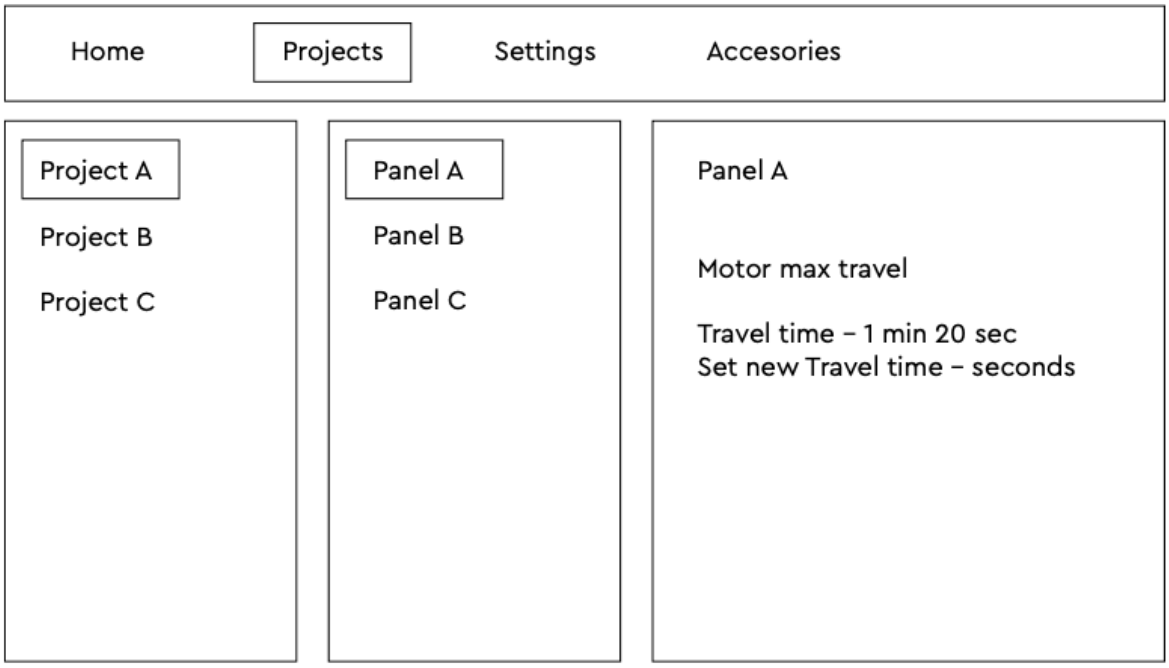
Use: settings/info.txt



2- PROJECTS

Use: projects folder

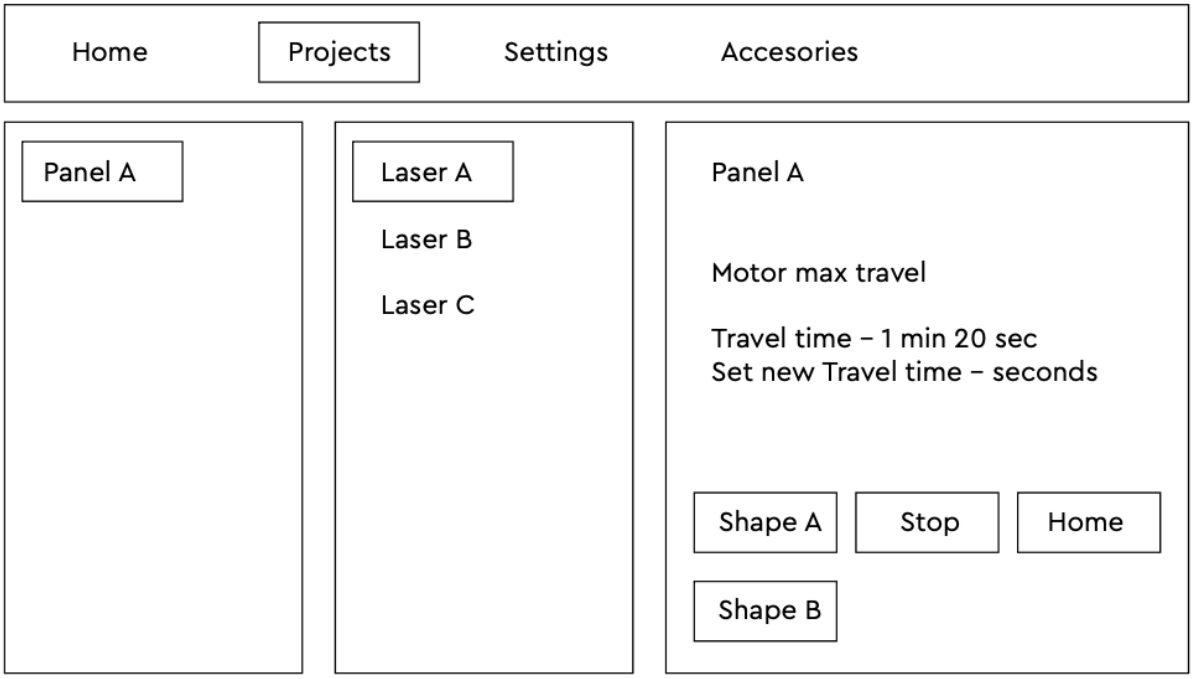
- On the left side, list the project folder names within projects folder. As a default: List the project from the newest to the oldest installed. (It can be changed from settings from the oldest to newest, and changed alphabetically listed.)
- On the middle, list the panel names of selected project folder. *(projects/Project A/…all of folder names…)*
- On the right side, show the information about selected panel. Take the information from panel A.txt file (location: projects/project A/panel A/panel A.txt). Show motor max travel value (find max value in panel A.txt). Show travel time for shaping and it can be set it up to new value.



2.1 Open the panel

1 *Use: projects folder, panel text files and settings/calibration/homePosition.txt*

- Show panel name on the left side.
- Show laser files on the middle. projects/ Project A/Panel A/ laserProjection/ .txt
- Show travel time and buttons on the right side.
- Show travel time and can be set it up to new value.
- Place 4 buttons
    - Shape A (selected panel)
    - Shape B (next panel)
    - Stop Button (to stop the shaping)
    - Home Button (to go to home position, value from homePosition.txt in calibration folder, in preferences.
- Make the buttons red color if there is an error on motors, and do not allow to click any button.

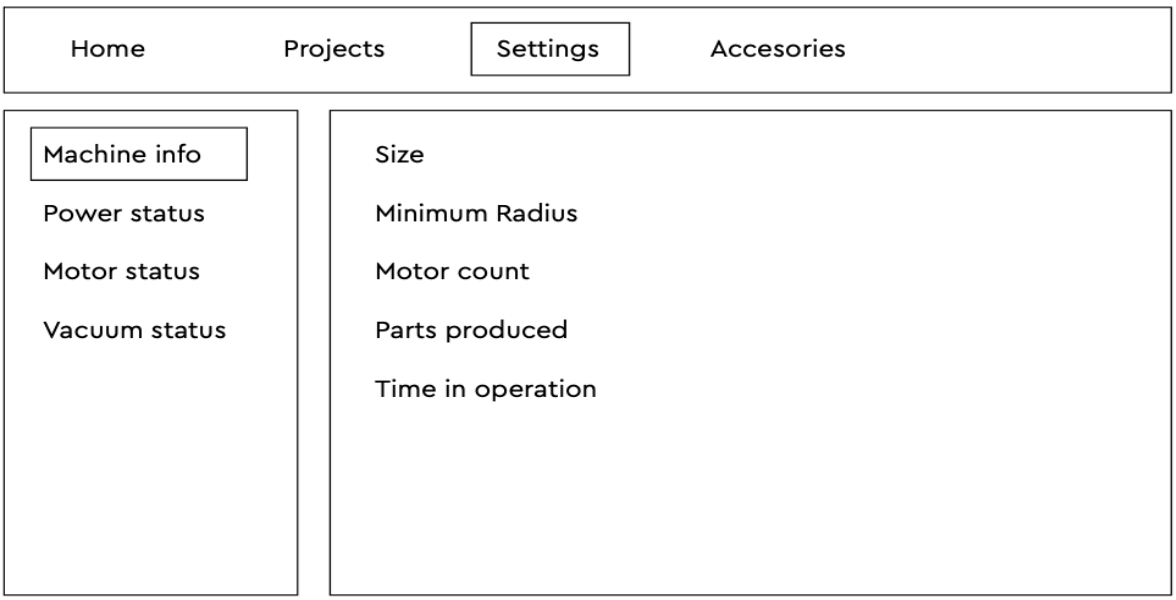


2.2 Click "START" (Button)

- Take the motor positions from Kurvatur Project A/Panel A.txt file to motors with travel time.
- Run the motors and go to the positions within travel time. Show its progress with progress bar.
- Update the status folder in preferences folder. Create a txt.file when clicked Shape buttons. Fill the part inside the file. (Check folder to see details)

3- SETTINGS

Use: settings/info.txt, *settings/ motor.txt, settings/errors.txt*

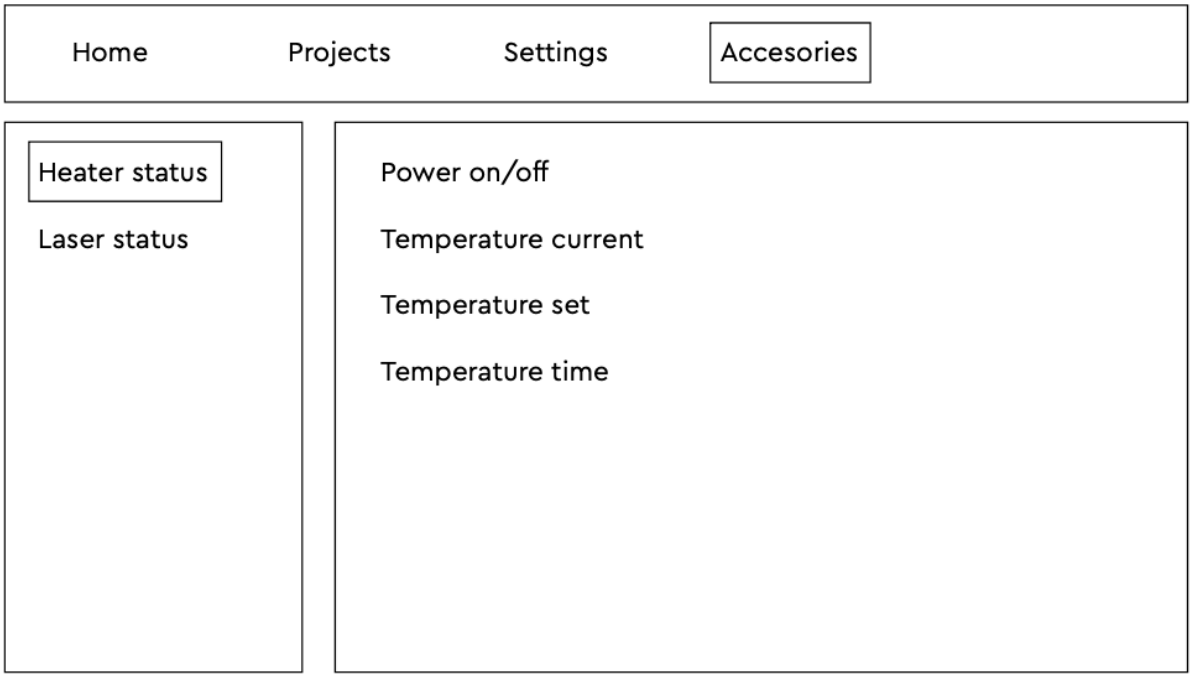


4- ACCESORIES

Use: settings folder

4.1 Heater Status

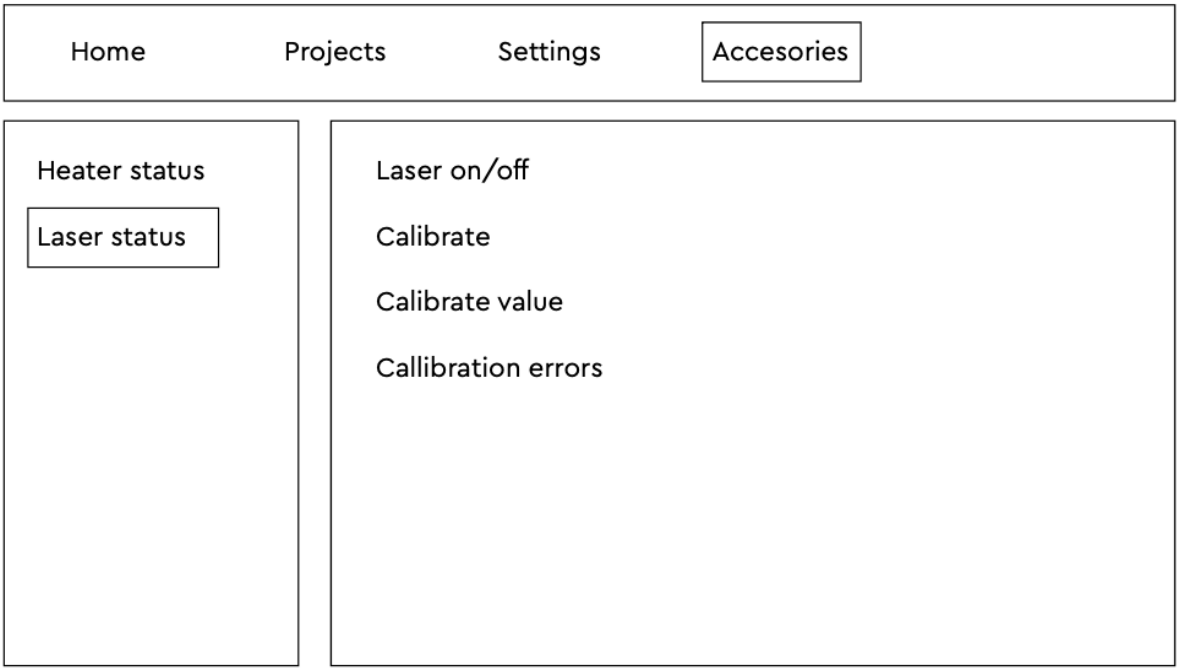
Use: settings/info.txt



4.2 Laser Status

Use: settings/calibration/laserPosition.txt, *settings/errors.txt*

- Place on/off button
- Place a calibrate button (when click, send the data to laser from laserPosition.txt)
- Show the errors if missing position



ERRORS

Use: settings/errors.txt

Write the errors over the file: the settings/errors.txt

Error Date: day/month/year
Error Time: hour.min.sec
Error Explanation:
Error Code/Name:

Errors:

1- Power Error (Show: “Motors were not powered”)
2- Motor Power Error (Show: “Motor number … was not powered”)
3- Motor Communication Error (Show: “Motor number … is not responding”)
4- Main Board Error (Show: “Main board not responding”)
5- Motor Number Repeating (Show: “Motor number … is repeating”)
6- Stalling Error (Show: “Motor number … stalled. It is delayed”)
7- Motor Invalid Error
8- Laser Calibration Error
9- Laser Temperature Error
10- Laser Invalid Error